

2D SCANNING LIDAR · LR-1BS SERIES

DUAL-OUTPUT OLEI LR-1BS5-V2 270° LIDAR

Zone logic and a mapping-grade point cloud from the same sensor: 16 banks of three levels with four switching inputs and four outputs, alongside 25 metres of range and $< \pm 20$ mm accuracy on common targets.



■ WHY THIS SENSOR

- **Data and I/O from one sensor** — 16 banks $\times$ 3 levels and 4 in / 4 out, while the point cloud keeps streaming
- **0.2–25 m at 80% reflectivity** — 2.5 $\times$ the LR-1BS2 envelope, on the same mounting footprint
- **$< \pm 20$ mm absolute accuracy on common targets** — $< \pm 40$ mm on reflectors, < 20 mm repeat — tight enough to localise on
- **User-settable weak-signal threshold** — raise it in a dusty bay, lower it for dark targets
- **Built-in multi-level filtering** — configurable tailing suppression before the cloud leaves the sensor
- **130 ms response at 10 Hz, 0.225° fixed resolution** — 16K–40K points per second over 100 Mbps Ethernet
- **Oforge zone configuration plus OLAMViewer and ROS** — draw the banks in the GUI, stream the cloud into the stack
- **450 g, IP65, 5 W at 12–24 V DC** — M8 power and LAN on 2 m leads, bottom or back mounting

0.2–25 m

WORKING RANGE

16 $\times$ 3

ZONES / LEVELS

4 / 4

INPUTS / OUTPUTS

$< \pm 20$ mm

ABSOLUTE ACCURACY

■ APPLICATION SCENARIOS



AMR & AGV

270° obstacle avoidance from a single chassis-mounted scanner



SLAM NAVIGATION

Point-cloud mapping and localisation for mobile platforms



INTRALOGISTICS

Forklifts, tuggers and conveyor zones in active aisles



INDUSTRIAL CELLS

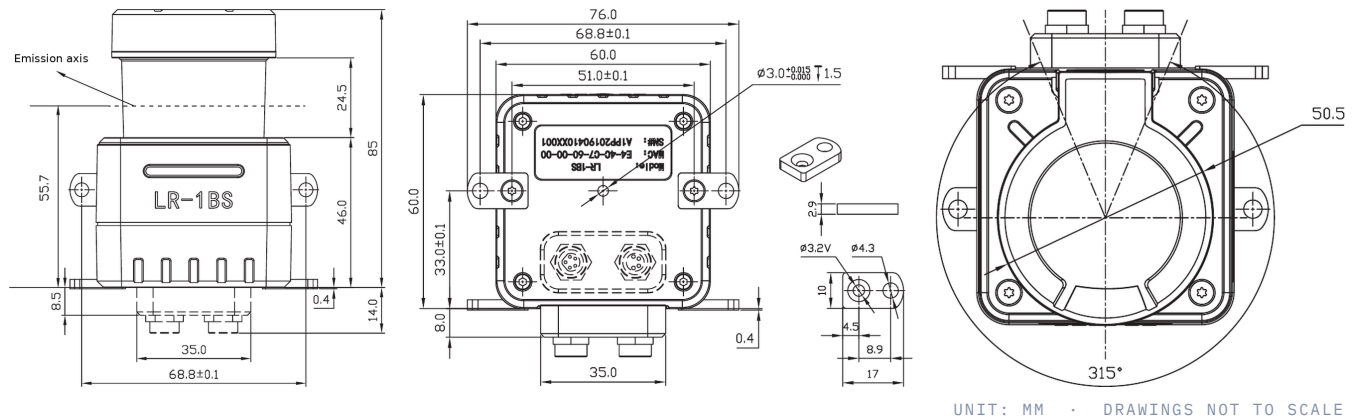
Machine-envelope monitoring and presence sensing

LR-1BS5-V2 — SPECIFICATIONS

PARAMETER	TECHNICAL DATA
Sensing technology	2D scanning ToF LiDAR
Laser wavelength	905 ± 20 nm
Laser safety level	Class 1 (IEC 60825-1:2014)
Laser channels	1
Field of view (horizontal)	270°
Beam divergence	5.8 mrad longitudinal 0.58 mrad transversal
Angle resolution (horizontal)	0.225°
Scanning frequency	10–25 Hz
Working distance (based on reflectivity)	0.2–25 m @ 80% reflectivity 0.2–8 m @ 10% · 0.2–2 m @ 4% · 0.2–25 m @ reflector
Accuracy ①	Common targets: absolute < ±20 mm Reflector: absolute < ±40 mm · Repeat < 20 mm
Distance resolution	1 mm
Signal strength	0–60000
Scanning rate	16K–40K pts/s
Horizontal error (horizontal angle)	≤ ±1°
Response time	130 ms @ 10 Hz

PARAMETER	TECHNICAL DATA
Number of bank fields	16 banks, 3 levels/bank
Switching inputs	4
Switching outputs	4
Ethernet connection	100 Mbps Ethernet
Connection	5-pin M8×1 female, 2 m (power) 4-pin M8×1 female, 2 m LAN
Operating voltage	12–24 V DC
Power consumption @ 25 °C	5 W
Mechanical dimensions	≤ 60 × 60 × 85 mm
Weight	450 g
Degree of protection	IP65 or optional
Operating temperature	–10 to +50 °C
Storage temperature	–30 to +70 °C
Ambient light limit	80 000 lux
Indicator status	RGB × 4
Mounting	Bottom / back
Function software	Oforge
Basic software	OLAMViewer · ROS driver

■ OUTLINE AND MOUNTING



Certification. FCC, EMC, CE, RoHS, REACH, FDA. EMC per IEC 61000-6-2:2016-08 and IEC 61000-6-3:2006-07. Vibration per IEC 60068-2-6:2007. Shock per IEC 60068-2-27:2008.

Functional safety. The manufacturer's specification for this product states laser safety Class 1 per IEC 60825-1:2014 and the EMC, vibration and shock standards listed above. It does not state an IEC 61496 Type rating or an ISO 13849 performance level. Treat the zone outputs as obstacle-avoidance and area-monitoring signals, not as a rated safety function, and do not use this sensor as the sole protective device. Where a rated scanner is required, ask MorpheusTEK about the OLEI GS1-5.

Notes. ① Standard value; actual value depends on environmental conditions. Range and accuracy both degrade with lower target reflectivity — size the zones from the 4% and 10% figures, not the 80% figure. Where 30 m of range matters more than the switching I/O, specify the LR-1BS5+; where a rated safety function is required, specify the GS1-5.

Installation. Mount the scanner so the 270° plane is unobstructed through its full sweep and the blind 90° faces structure, not traffic. Keep the optical window clear of spray and dust, clean it on a schedule in dirty environments, and verify every configured bank on the platform after any change to mounting height or angle.

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